

Mihir Ravindra Adelkar

United States | adelkar.mihir@gmail.com | (617)487-2363 | linkedin.com/in/mihir-adelkar | github.com/mihiradelkar

Dear Hiring Team,

I'm writing to express my strong interest in the [Role Title] position. With 3+ years building production healthcare AI systems, I bring hands-on experience in LLM integration, clinical data interoperability, and scalable architecture that directly addresses the challenges healthcare technology companies face today.

Most recently, I engineered a Prior Authorization AI system that transformed a 3-5 day manual review process into automated 5-second decisions with 95%+ accuracy. The system processes 100+ authorization requests daily, reducing clinician workload by 80

- RAG-based architecture combining patient data with coverage guidelines for intelligent decision-making
- FHIR compliant NLP pipeline extracting structured data (CPT codes, medications, diagnoses) from unstructured clinical notes
- Model Context Protocol (MCP) design enabling full audit trails, citation traceability, and regulatory compliance
- HIPAA-compliant PHI handling with encryption and comprehensive security controls

Beyond this project, I've architected full-stack systems serving 500K+ users, built distributed microservices with Kafka and gRPC, and established CI/CD pipelines achieving 85% test coverage. My technical foundation spans Python/FastAPI, React/TypeScript, Node.js, AWS/Kubernetes, and modern observability tools.

What differentiates my background is the combination of healthcare domain expertise (FHIR/HL7 interoperability, clinical workflows, regulatory requirements) with production AI/ML systems experience—a skill set that's essential but rare in this space.

Thank you for your consideration.

Best regards,
Mihir Ravindra Adelkar